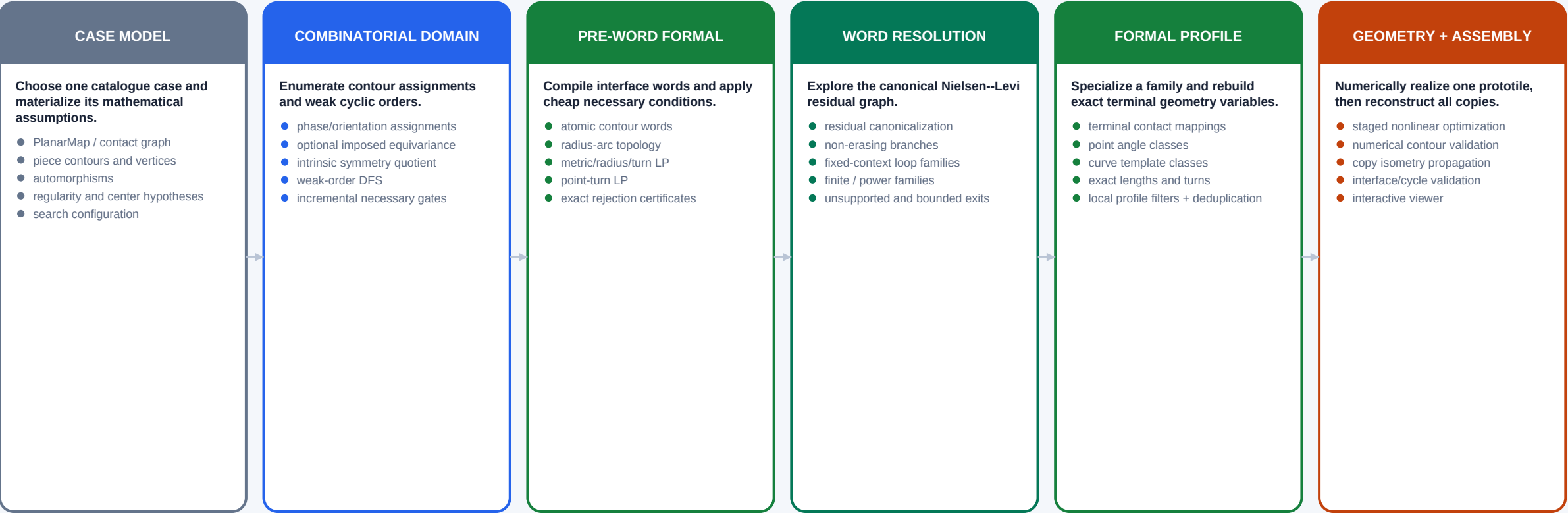


1. System overview

One candidate moves left to right; side exits classify why it does not continue.

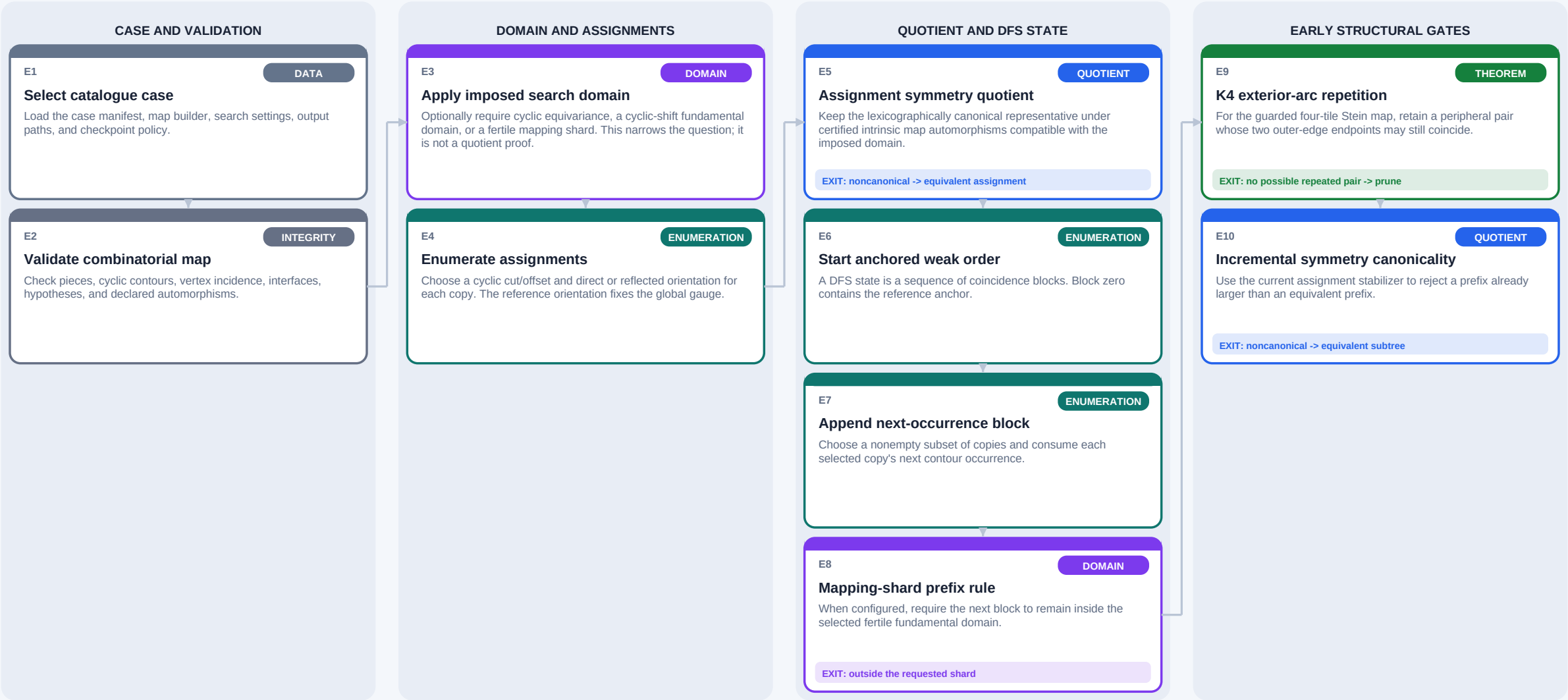


MAPPING LAB (EXPERIMENTAL SIDECAR)

Samples and ranks complete mappings, then evaluates them through a production-filter adapter. Learning changes proposal order only; it does not change production acceptance rules.

2. Catalogue, assignments, and weak-order construction

Blue nodes remove equivalent representatives; purple nodes intentionally restrict the question being searched.

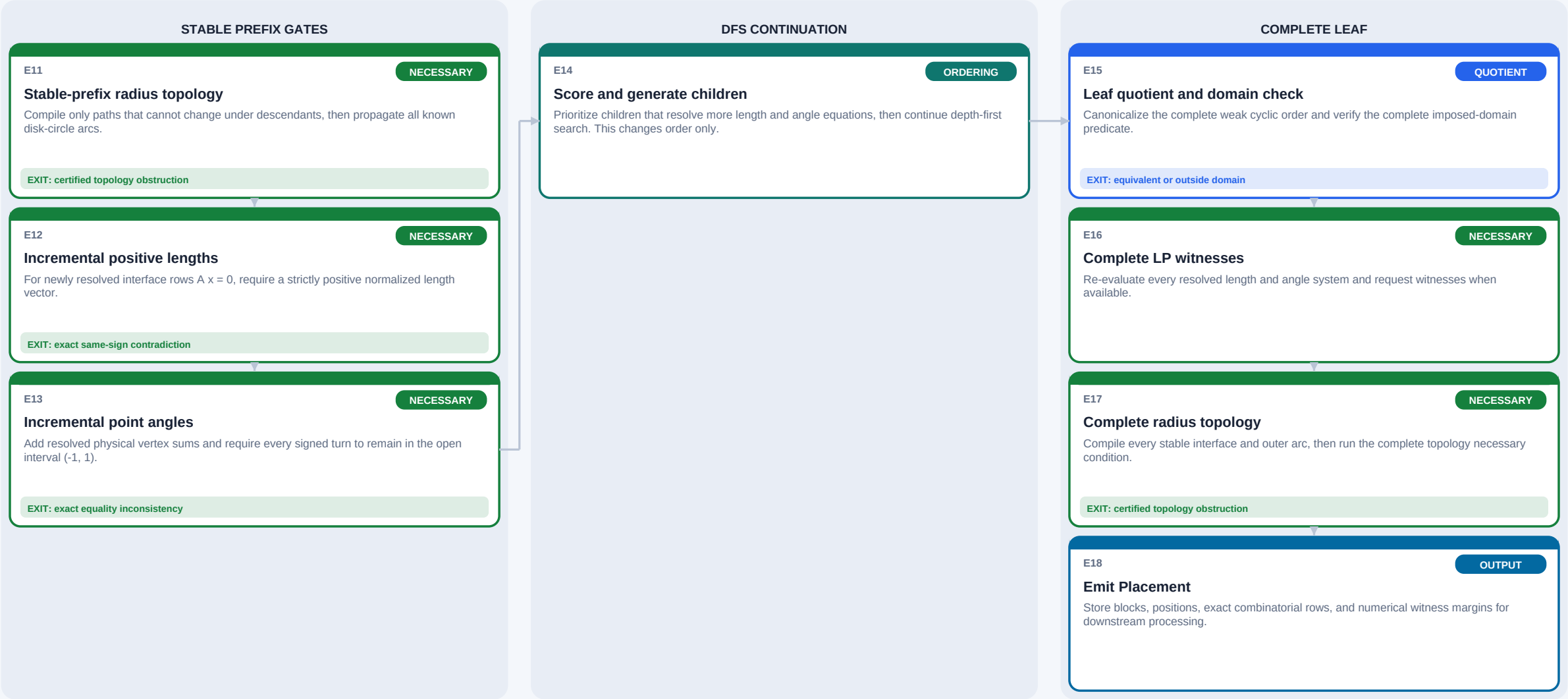


The imposed domain and the intrinsic quotient interact, but have different meanings: one changes the requested search space; the other only selects a representative.

● equivalence quotient ● imposed domain ● necessary condition ● deferred / bounded ● numerical non-proof ● surviving output

3. Weak-order prefix gates and complete Placement emission

Only stable information is screened at a prefix; all complete gates are rerun at the leaf.

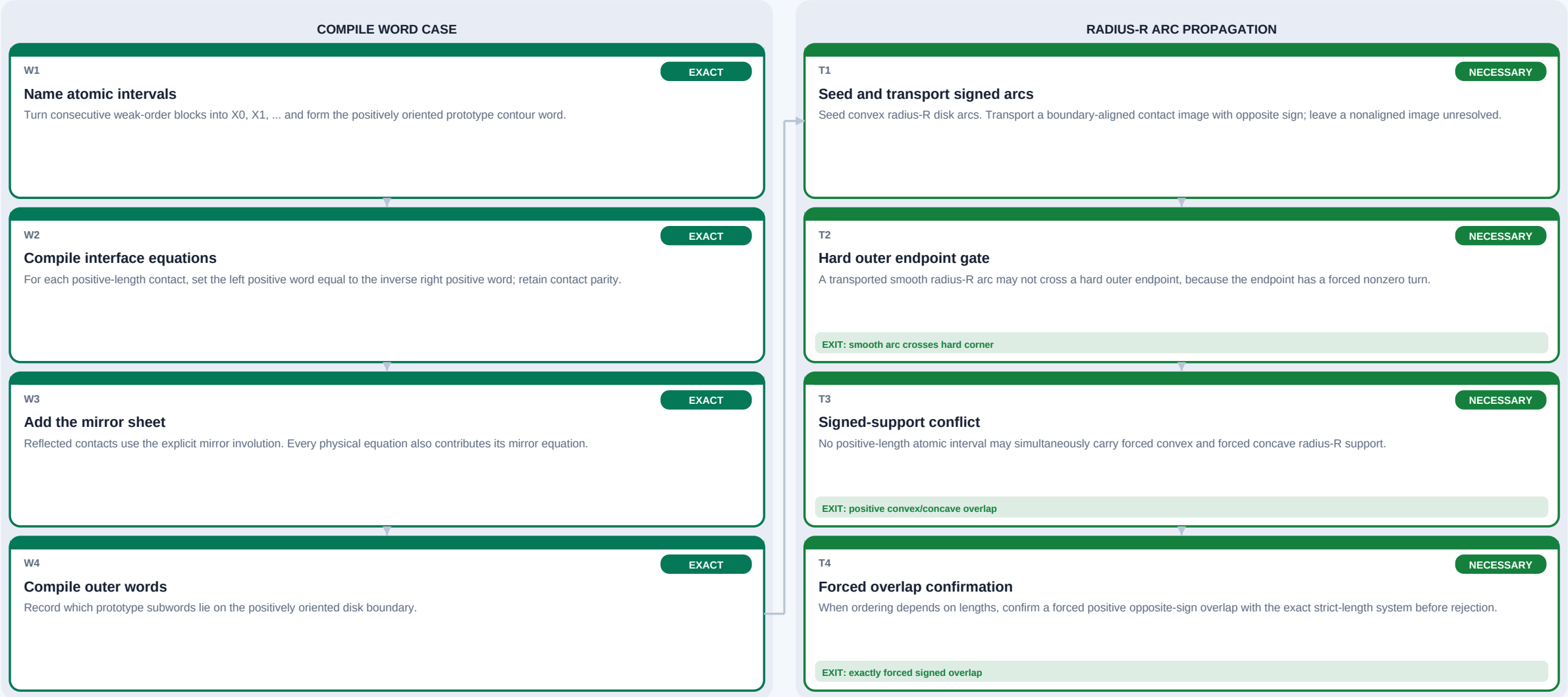


Special cyclic-shift mode constructs one fundamental half/order and its images directly; it still applies every complete leaf gate before emission.

● equivalence quotient ● imposed domain ● necessary condition ● deferred / bounded ● numerical non-proof ● surviving output

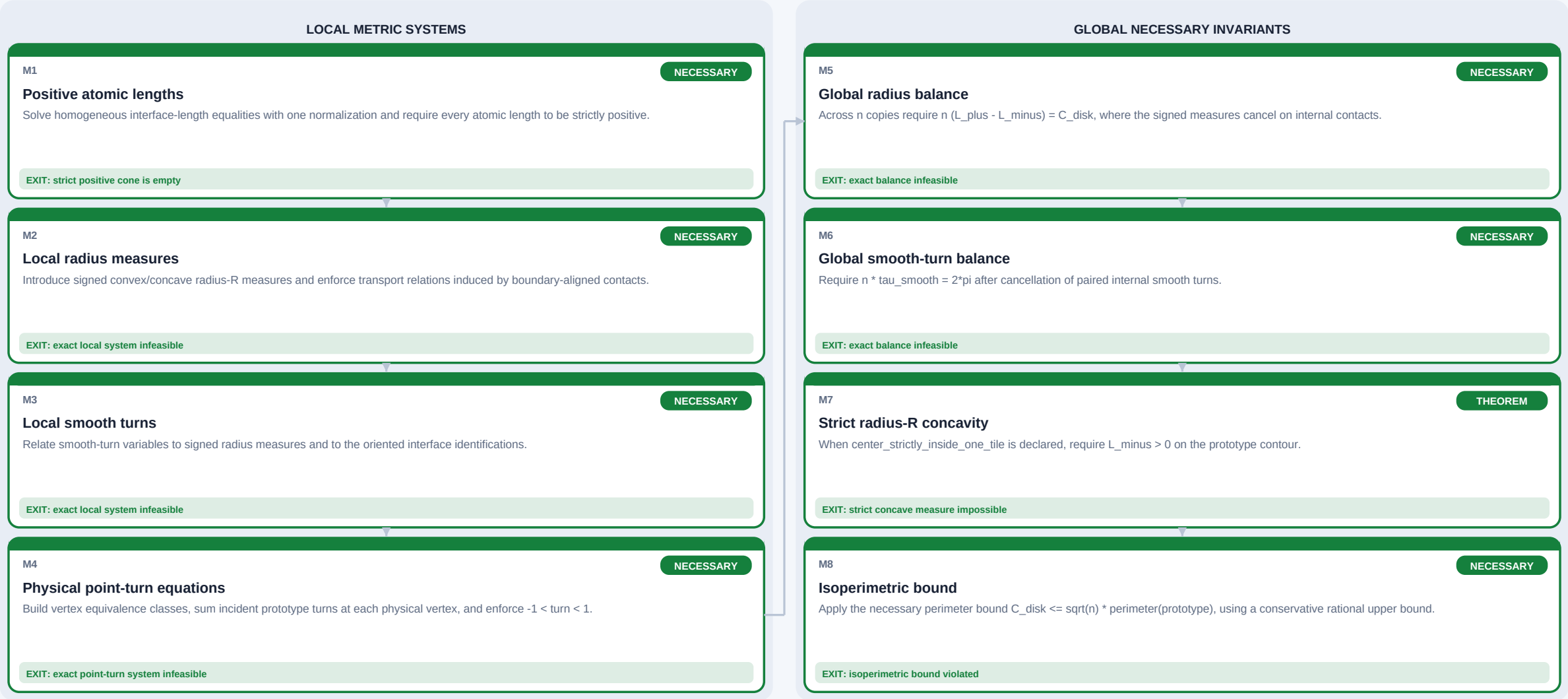
4. Word compilation and radius-arc topology pruning

Word equations are exact; green topology exits are intended necessary-condition certificates.



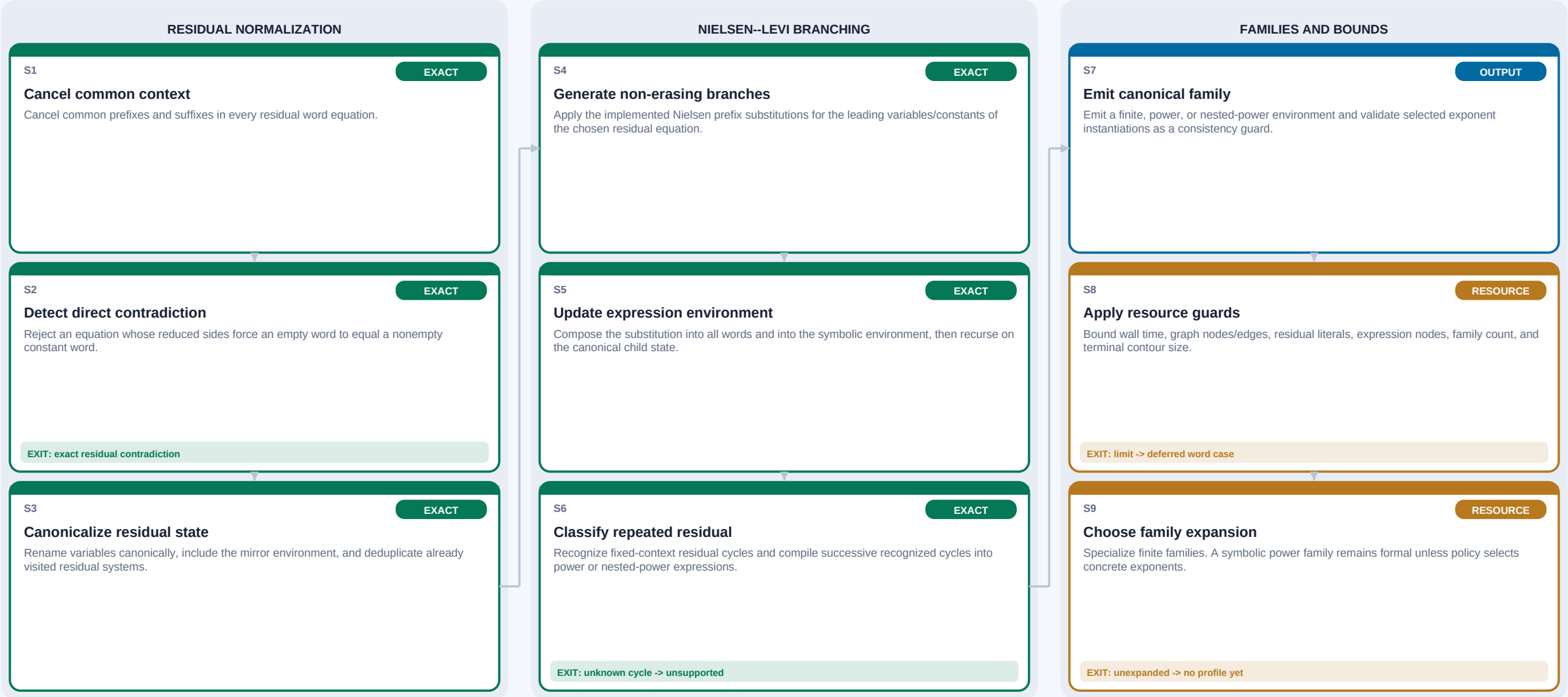
5. Exact pre-word metric and turn curriculum

The systems are nested in this order; a rejection requires an exact rational contradiction or nonpositive strict margin.



6. Exact-partial word resolution and family policy

Amber exits mean unresolved or unsupported, never a proof of impossibility.

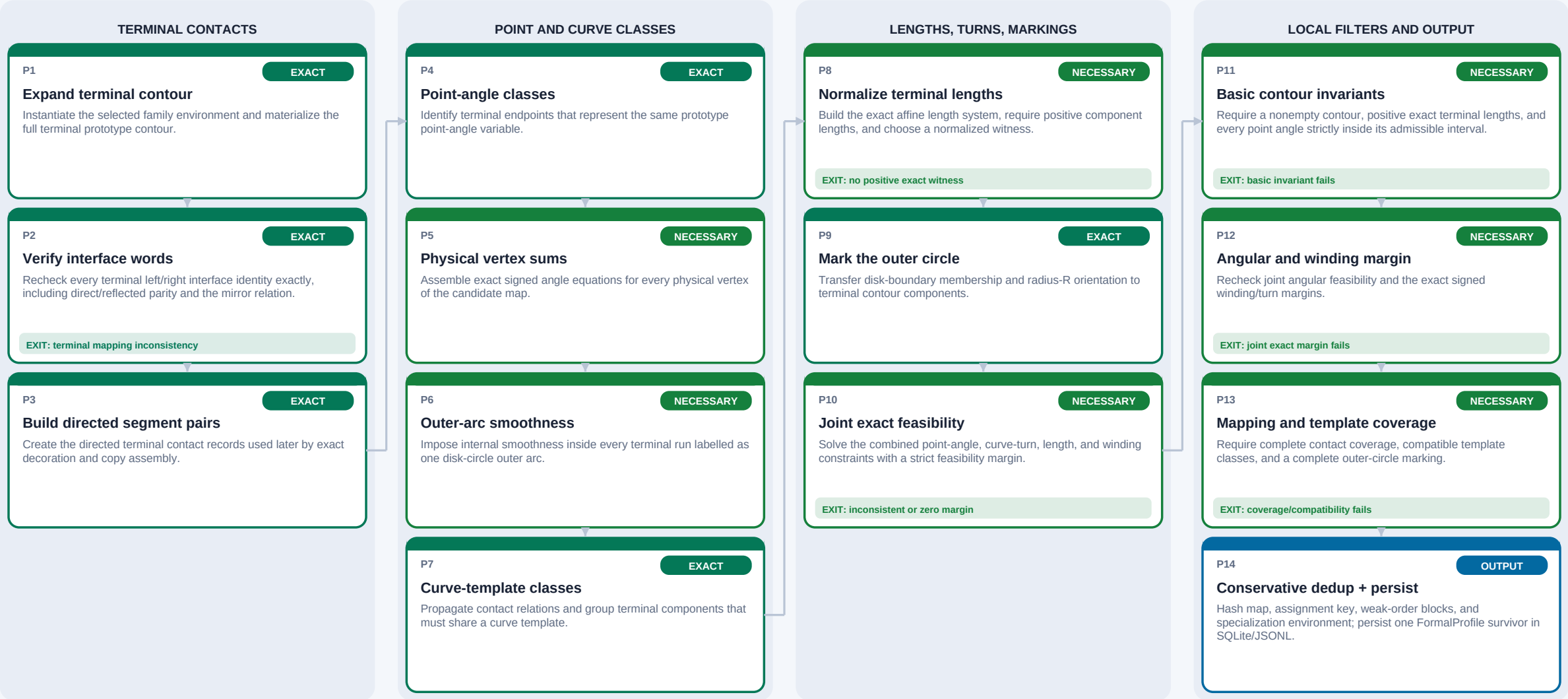


Compile or solver exceptions taint the campaign and may skip the affected work item. They are integrity failures recorded in errors.jsonl, not rejections.

● equivalence quotient ● imposed domain ● necessary condition ● deferred / bounded ● numerical non-proof ● surviving output

7. Terminal contact system, exact decorations, and profile filters

A specialized word family is rebuilt as an exact formal geometry profile before any numerical search.

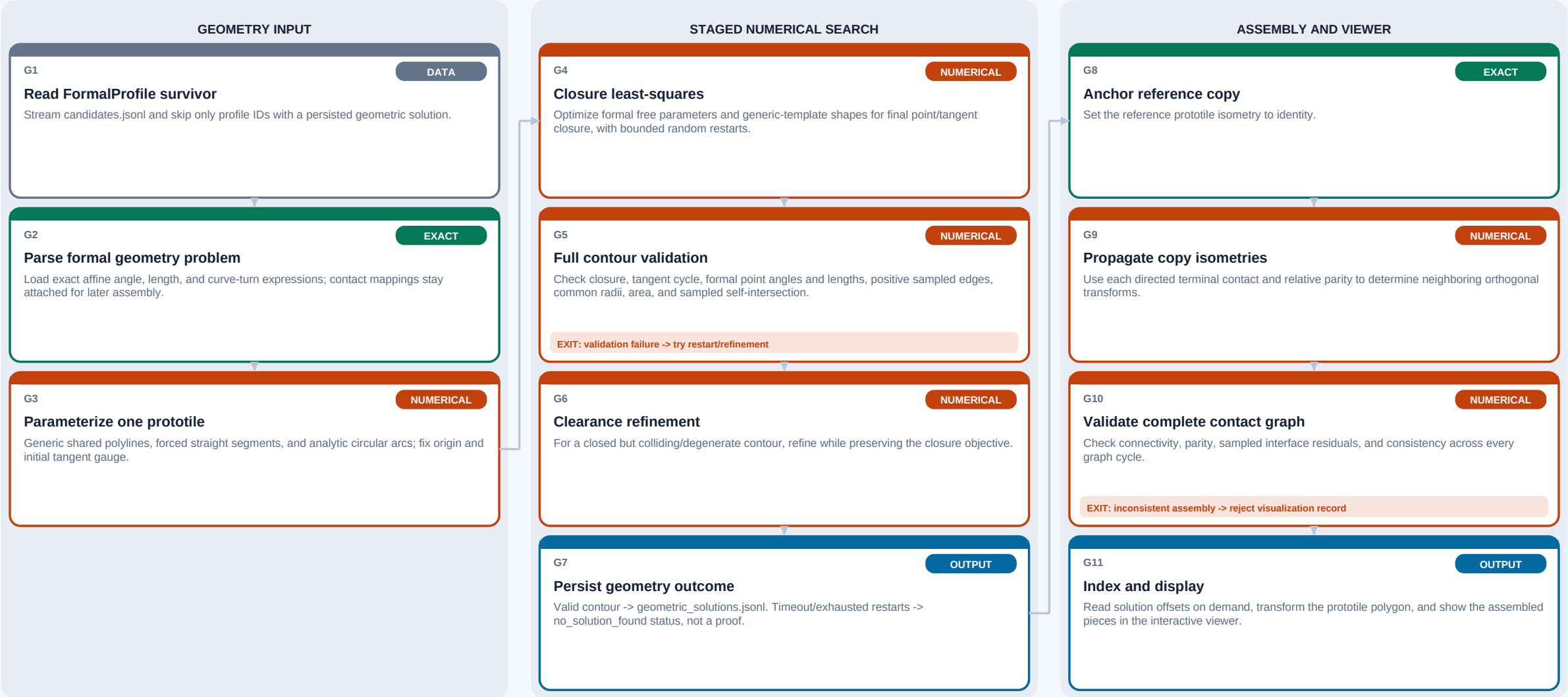


Profile-build exceptions taint the campaign. The optional cyclic_no_backtracking heuristic is disabled by default because it has a known counterexample.

● equivalence quotient ● imposed domain ● necessary condition ● deferred / bounded ● numerical non-proof ● surviving output

8. Numerical prototile geometry and mapping-derived assembly

Orange outcomes are numerical evidence only; absence of a solution is not a formal impossibility result.



The current geometry stage solves a single piece first. It does not optimize all copy placements jointly; assembly is reconstructed deterministically afterward from formal contacts.

● equivalence quotient ● imposed domain ● necessary condition ● deferred / bounded ● numerical non-proof ● surviving output

9. Meaning of exits and open mathematical provenance

Only the first two green/blue classes can support exhaustive formal conclusions.

Intrinsic quotient Removes a representative only when a certified automorphism maps it to another enumerated representative.	Imposed domain Answers a narrower question by construction (rotation/equivariance/fertile shard); not valid for the unrestricted problem.	Necessary-condition rejection Intended mathematical impossibility certificate. Floating screens are conservative; exact rational checks confirm sensitive LP rejections.	Deferred / unsupported A graph, expression, family, contour, or time budget ended the attempt. Must never be reported as unsatisfiable.	Numerical non-proof Optimization or sampled validation did not find a realization. This is evidence, not a formal rejection.	Integrity / implementation error Malformed or inconsistent constructed data, or an unexpected exception. Campaign is tainted when work may have been skipped.
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Q1	Transported exterior-arc repetition theorem For the guarded four-tile Stein map, why must there exist peripheral $i \neq j$ with the same oriented prototype outer interval, equivalently $\text{start}_i = \text{start}_j$ and $\text{end}_i = \text{end}_j$ in the weak cyclic order?	NEEDED: Exact theorem statement, all geometric regularity/center hypotheses, and the Kurusa--Langi--Vigh source or a self-contained proof.
Q2	Strict radius-R concavity in a Stein configuration Why does $\text{center_strictly_inside_one_tile}$ imply $L_{\text{minus}} = \sum_i r_i^{\wedge} > 0$ on the prototype contour, rather than merely $L_{\text{minus}} \geq 0$?	NEEDED: A proof that a positive-measure disk-circle arc is transported to an internal interface with opposite sign under exactly the hypotheses encoded by the map flag.
Q3	Completeness of Nielsen branching Why do $\text{branch_substitutions}$ enumerate every non-erasing solution of each residual word equation, modulo the canonical renaming/inversion used by the graph?	NEEDED: A standard Nielsen--Levi lemma/citation or a proof tied to the implemented branch cases.
Q4	Fixed-context residual-cycle compilation Why does a detected cycle $V \rightarrow p \vee s$ generate exactly the family $p^n \vee s^n$ for all $n \geq 1$, and why do successive compiled cycles preserve completeness for nested powers?	NEEDED: A soundness/completeness proof for <code>classify_fixed_context_loop</code> and the recursion that restarts from the ancestor residual.

WARNING: The optional `cyclic_no_backtracking` profile heuristic is disabled by default and has a known counterexample. If enabled, it is explicitly non-mathematical and must not support an exhaustive claim.

10. Implementation map

Primary production modules audited for this diagram.

Pipeline responsibility	Primary module(s)	Key entry point(s)
Catalogue / orchestration	orchestration/catalog.py, orchestration/pipeline.py	CaseCatalog, PipelineExecutor
Map and hypotheses	maps/base.py, maps/registry.py	PlanarMap, ProblemHypotheses, MapAutomorphism
Assignments / imposed symmetry	enumeration/assignments.py, mapping_subdomain.py	AssignmentEnumerator, MappingSubdomain
Intrinsic quotient	enumeration/symmetry.py	MappingSymmetryQuotient
Weak cyclic orders	enumeration/weak_orders.py	WeakOrderEnumerator
K4 theorem gate	enumeration/exterior_arc_repetition.py	ExteriorArcRepetitionConstraint
Early LP gates	constraints/length_lp.py, constraints/angle_lp.py	LengthFeasibilityOracle, AngleFeasibilityOracle
Radius topology	preword/prefix_topology.py, preword/arc_topology.py	PrefixRadiusArcTopologyFilter, RadiusArcTopologyFilter
Pre-word linear systems	preword/linear_invariants.py, preword/filter.py	PrewardLinearInvariantFilter, PrewardPruningPipeline
Word compilation	words/compile.py	compile_word_case, build_terminal_contact_system
Exact-partial words	words/compact.py, words/exact_partial.py	ExactPartialWordSolver
Family expansion	words/families.py	expand_family, FamilyExpansionPolicy
Terminal decoration	profiles/build.py, profiles/decorations.py	build_formal_profile, build_decorations
Local profile filters	profiles/filters.py, profiles/canonical.py	ProfileFilterPipeline, conservative_profile_key
Formal search driver	pipeline/runner.py	SolverRunner
Numerical geometry	geometry/model.py, geometry/solver.py, geometry/validation.py	NumericalContourSolver, validate_geometry
Assembly / display	visualization/assembly.py, visualization/validate.py, visualization/runner.py	assemble_geometric_solution, VisualizationRunner
Mapping Lab sidecar	mapping_lab/core.py, deep_experiment.py	CompleteMappingEvaluator, DeepExperimentRunner